ZHAN'S GOURMET GOODIES

Bakery Process Data Modeling

LEAD CONSULTANTS

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PART 1: Database Design

Team composition:

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Scope:

Zhan's Gourmet Goodies Bakery stands at the forefront of culinary excellence, with a chain of eight esteemed bakeries dotted across California. Known for its scrumptious cookies, cakes, pastries, and bread, Zhan's Gourmet Goodies prides itself on both the quality of its baked delights and the efficiency of its operations. At the heart of their logistical prowess are multiple central warehouses that act as the nexus for product distribution, ensuring that each bakery is stocked with the freshest ingredients to craft their artisanal goods.

In response to a surge in demand, Zhan's bakery is poised to refine their inventory management system, a strategic move designed to bolster their supply chain and inventory control. The goal is to enhance the ability to manage stock across all outlets, maintain a robust supply of diverse ingredients, and ensure the highest standards of product deliveries.

The upcoming Enterprise Resource (ER) assignment will delve into the intricacies of Zhan's Gourmet Goodies' inventory management system. It will establish a comprehensive database that encapsulates the multifaceted relationships among products, ingredients, suppliers, procurement, warehouses, bakeries, and employees. This database will serve as the blueprint for orchestrating inventory levels, tracking inventory, and analyzing bakery demand patterns. By meticulously defining the entities and attributes within this domain, the assignment will lay the groundwork for Zhan's Gourmet Goodies to scale operations, reduce waste, and continue delivering exceptional baked goods to their customers throughout California.

Business Rules:

Zhan's Gourmet Goodies needs to efficiently manage the inventory for each product and ingredient across multiple bakery locations. Ensure a steady supply of ingredients from various suppliers. Track inventory procurement. Monitor the quantity, and expiration dates of ingredients. As part of their inventory management system, they aim to create a database that not only records inventory levels but also establishes relationships between products, ingredients, suppliers, procurement, deliveries, and bakery locations.

- Make a ternary relationship between location, product, and driver (BR: one driver delivers many products to many locations)
- Ternary relationship between manager, ingredients, and supplier (BR: Manager signs on delivery of ingredients coming from the supplier and that's called Procurement.)

- Employee subtypes have partial specialization: (BR: managers manage many other types of employees and in other types of employees we want to only track information about our bakers and drivers.)
- Binary relationship between baker and ingredient (Many bakers expend many ingredients, we need to track as they expend the ingredients.)
- Binary relationship between ingredients, and product (BR: Many ingredients are expended to make many products and we need to track this as recipe)

This will allow them to make data-driven decisions, streamline their supply chain, and provide a consistent and delightful bakery experience to their customers. By implementing these entities and their relationships, Zhan's Gourmet Goodies can maintain their reputation for quality and expand their bakery business to meet the growing demand.

Definitions:

Entity	Attribute
Product	Represents the bakery products available for sale, including cookies, cakes, pastries, and bread. Includes attributes such as product name, product ID (primary identifier), description, price, and category.
Ingredient	Represents the raw materials used in baking. Includes attributes like ingredient ID (primary identifier), ingredient name, supplier ID, unit of measurement, quantity on hand, price, and expiration date (if applicable).
Supplier	Represents the suppliers providing ingredients to your bakery. Includes attributes such as supplier name, contact information, and supplier ID (primary identifier).
Procurement	Tracks all inventory-related transactions, including receiving ingredients, stocking products, and adjusting quantities. Includes attributes such as transaction ID (primary identifier), date, type (e.g., receipt, sale, adjustment), product/ingredient ID (foreign identifiers), quantity, and location.
Location	Represents the physical locations where inventory is stored, including the central warehouse and individual bakery locations. Attributes may include location ID (primary identifier), location name, and address.
Employee	Employees that work in the warehouse. They are uniquely identified by EmployeeID (primary identifier). The attributes of these employees include name, last name, address, and type of employee (Manager or Other).
Employee_Manager	Manager is a type of employee that manages Other employees. The primary identifier is EmployeeID.

Employee_Other	Other is a type of employee that is managed by Manager employees. The primary identifier is EmployeeID.
Employee_Other_Baker	Employee_Other_Baker is an employee who is managed by a manager and is specifically a baker who utilizes ingredients in the warehouse to come up with new products. The primary identifier is EmployeeID which is a foreign key from Employee_Other.
Employee_Other_Driver	Employee_Other_Driver is an employee who is managed by a manager and is specifically a driver who makes deliveries to the bakeries. The primary identifier is EmployeeID which is a foreign key from Employee_Other.
Location_Bakery	Represents the location of the bakery. This is a type of location where the primary identifier is LocationID.
Location_Industrial_Kitchen	Represents the location of the industrial kitchen. This is a type of location where the primary identifier is LocationID.

E-R Diagram: